

Analysis Report

1. Executive Summary

The analysis followed a single narrative arc from diagnosis to real-world glucose behavior, to meal response, to prediction across 45 participants (15 Normal, 16 Prediabetes, 14 Type 2 Diabetes). Continuous glucose monitoring showed that average glucose, variability, and post-meal burden worsened progressively across the three groups. Prediabetes was not a uniform category: participants fell along a Normal-like to T2D-like continuum rather than into two clean subgroups. The strongest predictive model for post-meal glucose response combined meal composition, individual metabolic profile, and post-meal activity; gut/microbiome summary features did not add measurable predictive value in this exploratory sample.

Central question: “How much of the glucose response comes from the meal, and how much comes from the person?” “Why Similar Meals Create Different Glucose Responses” These questions anchored every hypothesis test, model comparison, and case study in the project.

2. Dataset Overview

The CGMacros dataset is unusual in that it combines continuous glucose monitoring, detailed meal logs, activity tracking, clinical biomarkers, and gut microbiome functional scores for the same 45 participants enabling questions that most single-source datasets cannot answer. Each participant contributes roughly ten days of continuous data.

- 45 participant-level CGM/fitness files (1-minute interval readings) plus meal metadata: calories, carbohydrate, protein, fat, fiber, meal type, and photo path.
- Three supplementary files: bio.csv (demographics, anthropometrics, blood analytics, finger-stick glucose), microbes.csv (1979 bacteria presence/absence indicators), and gut_health_test.csv (22 ordinal gut-health scores).
- Participants classified by HbA1c into Normal (<5.7%), Prediabetes (5.7–6.4%), and Type 2 Diabetes (>6.4%).

3. Key Insights

3.1 Glucose Behavior Worsens Progressively Across Groups

Subject-level CGM profiling (average glucose, standard deviation, coefficient of variation, time in/above/below range) moved the analysis beyond a static diagnosis and showed how glucose behaved continuously in daily life.

Metric	Normal	Prediabetes	T2D	Interpretation
Average glucose (mg/dL)	84.70	105.16	137.62	Progressively higher glucose burden

Metric	Normal	Prediabetes	T2D	Interpretation
Glucose variability	18.40	23.82	38.25	Greater instability with worsening status
Time above 180 mg/dL (%)	0.65	2.66	16.54	Marked rise in hyperglycemic exposure
Time below 70 mg/dL (%)	23.09	8.53	1.50	Standard TIR alone was misleading for the Normal group

Sensor validation was a necessary first step: very low Libre readings were checked against Dexcom, and while correlation with HbA1c was similar across sensors, Libre aligned slightly better with fasting glucose and was retained as the primary CGM signal. High correlation between sensors did not always mean close agreement an important distinction for downstream analysis.

3.2 Prediabetes Behaves as a Continuum, not a Category

The 16 Prediabetes participants showed wide metabolic variation across BMI, fasting glucose, insulin, TG/HDL ratio, average glucose, and glucose variability. K-means clustering suggested two clusters, but the split was highly unbalanced (14 vs. 2) and was not reproduced by hierarchical clustering rather than overstate a weak cluster structure, the analysis shifted to a continuous metabolic similarity score.

Metabolic similarity score: each Prediabetes participant was positioned between typical Normal and T2D profiles (negative = more Normal-like, positive = more T2D-like). This score correlated with independent CGM measures not used to construct it $r = 0.724$ with time above 140 mg/dL and $r = 0.615$ with time above 180 mg/dL supporting Prediabetes as a continuum rather than a single homogeneous state.

3.3 Meal Response and Metabolic Flexibility Worsen with Status

From 1706 recorded meals, 1698 complete 2-hour response windows were built, capturing baseline glucose, glucose at 30/60/90/120 minutes, peak rise, time to peak, 2-hour incremental area under the curve (iAUC), and recovery. iAUC and the 120-minute recovery value describe metabolic flexibility more completely than peak glucose alone.

Outcome	Normal	Prediabetes	T2D
Mean peak rise (mg/dL)	28.19	41.68	58.95
Mean 2-hour iAUC	1627	2634	3773
Glucose above baseline at 120 min (mg/dL)	9.19	19.56	32.10
Mean time to peak (min)	56.96	62.31	71.19

Mixed-effects models, which accounted for repeated meals within the same person, confirmed these differences statistically: relative to Normal, Prediabetes showed a +12.35 mg/dL higher peak rise ($p = .020$), +906 higher iAUC ($p = .041$), and +10.35 mg/dL greater 120-minute glucose change ($p = .009$),

with larger differences for T2D. Critically, metabolic status remained significant even after adjusting for carbohydrates, protein, fat, fiber, and meal type meal composition alone did not explain the response.

3.4 Predictive Modeling: The Person Matters as Much as the Meal

Random Forest models were evaluated with GroupKFold cross-validation by subject, preventing meals from the same person from appearing in both training and test folds.

Model	Peak Rise R ²	iAUC R ²	Interpretation
Meal only	0.159	0.129	Meal contains useful signal
Person only	−0.017	−0.065	Person features alone did not generalize well
Meal + Person	0.339	0.263	Large improvement
Meal + Person + Activity	0.376	0.312	Best-performing model; modest activity gain
+ Gut/Microbiome	0.319	0.286	No additional improvement in exploratory subset

Activity data (METs and Intensity, recorded for complementary, non-overlapping participant groups) was harmonized by standardizing each within its own measurement system and combining them into a unified relative activity-intensity score, covering 1,693 of 1,698 meals across all 45 subjects. The top cross-validated predictors were fasting glucose (0.458), carbohydrates (0.238), meal type (0.119), baseline glucose (0.095), and diabetes group (0.070) reinforcing that both the meal and the person's metabolic condition drive the post-meal response.

3.5 Matched-Meal Case Study: Identical Meals, Different Outcomes

The team searched for different participants who ate the same meal type with exactly matched recorded carbohydrate, protein, and fat content, requiring 100% consumption and no other recorded meal in the following two hours. This produced 9429 high-quality, exact macro-matched cross-group pairs. One illustrative pair, a T2D participant and a Normal participant who each ate a lunch of 94 g carbohydrate, 12 g protein, and 13 g fat showed dramatically different post-meal glucose trajectories, underscoring that identical recorded macronutrient composition does not guarantee an identical glucose response. This is presented as an observational illustration, not causal proof.

4. Data Preparation and Cleaning Insights

The throughline across every dataset was evidence before action: profile first, change only what a specific finding justifies, verify immediately after each change, and default to flagging or nulling over guessing or deleting.

4.1 Missingness Meant Different Things in Different Files

- CGMacros time-series: missingness was expected driven by differing sensor sampling rates and meal/activity fields only populating when an event occurred.

- Microbes: exactly 1979 missing values, all belonging to a single participant's (Subject 48) entire profile confirmed structural, not scattered, because the missing count exactly equaled the total column count.
- Gut Health Test: every one of the 22 score column was missing exactly 5 values (10.64%) an identical count across columns signaled a block pattern (a few subjects with no assessment at all) rather than random gaps.

4.2 Out-of-Range Values Required Case-by-Case Judgment

“Out of range” did not automatically mean “wrong.” The evidence used to decide differed by case:

- Libre glucose reading of 405 mg/dL (documented max: 400) was corrected because neighboring anchor readings (391 and 386 mg/dL) made the physiological trajectory clear, allowing linear interpolation to restore a continuous curve.
- METs = 0 (valid range 10–176) was set to ‘NaN’ because it co-occurred with missing heart rate and zero activity calories a pattern pointing to a device gap, not a true zero reading.
- Nutrition outliers (131 amount-consumed values over 100%, protein below 3g in snacks) were mostly kept and flagged rather than corrected, since valid and invalid values coexisted within the same participant files with no reliable way to infer a replacement value.

4.3 Some Extreme Values Were Error Codes, Not Measurements

In the Bio dataset, Subject 12 had an LDL of 800 and VLDL of 400 values so round and extreme they were recognized as lab error codes rather than physiological readings. These two values alone inflated the LDL standard deviation to roughly 110 and pushed the mean cholesterol/HDL ratio to 12.66 against a median of about 3, showing how a single record can distort dataset-wide statistics if left uncorrected.

4.4 Standardization Can Silently Destroy Information

Standardizing the 1979 microbiome column names collapsed two genuinely different organism names into one identical string (*ruminococcus_gnavus_atcc_29149*). This was only caught because the team explicitly checked for post-standardization uniqueness rather than assuming the transformation was safe a reminder to verify every automated cleaning step, not just the ones expected to be risky.

5. Challenges Faced and How They Were Overcome

Challenge	Resolution
45 separate participant files with unverified structural consistency	Compared row/column structure across all files before merging, rather than assuming uniformity
No subject ID embedded in the CGM files themselves	Extracted subject ID from filename and added a <code>source_file</code> column to every row for traceability
Mixed timestamp text formats across datasets	Converted with <code>datetime</code> (<code>errors='coerce'</code> for Bio/CGMacros) and explicitly verified a 100% success rate

Challenge	Resolution
Deciding whether to correct or flag out-of-range values	Established a rule: correct only with independent supporting evidence (e.g., neighboring readings); otherwise, flag and preserve the original
Distinguishing genuine device/sensor gaps from real zero-readings (METs)	Cross-referenced against co-occurring fields (HR, activity calories) instead of judging the range violation in isolation
Column standardization created an accidental name collision	Checked for post-standardization uniqueness and appended a disambiguating suffix rather than silently overwriting a column.
Distinguishing “no data collected” from “confirmed zero” in Microbes	Chose not to impute Subject 48's missing values with 0, since 0 carries a specific meaning (confirmed absence) that doesn't apply to unmeasured data
Random-looking vs. structural missingness in Gut Health	Used the uniform 5-missing-per-column pattern as evidence of a block gap, then applied drop (how='all') restricted to metric columns so no subject with even one real score was dropped
Physiologically impossible values that could be data-entry error or genuine extreme case	Investigated pattern, frequency, and plausibility per variable — keeping unusually high carb/fiber values as real dietary outliers while nulling the Bio lab error codes

6. Conclusion

The story progresses logically: diagnosis identified broad risk groups. CGM data showed real-world deterioration from Normal to Prediabetes to T2D. Prediabetes proved heterogeneous and was better represented as a continuum. Meal-level analysis showed larger spikes, greater 2-hour glucose burden, and poorer recovery with worsening metabolic status. Adjusted models confirmed these differences were not explained by meal composition alone and predictive modeling showed the strongest performance came from combining meal composition, individual metabolic profile, and post-meal activity.

Main takeaway: Post-meal glucose response is personalized. The key question is not only “What did the person eat?” but also “What metabolic state did that meal enter?”

External context: The Diabetes Prevention Program reported that intensive lifestyle intervention meaningfully reduced progression to Type 2 diabetes among high-risk adults, reinforcing that personalized, lifestyle-based strategies informed by analyses like this one have real-world prevention value.